Physics-Infused Reduced-Order Modeling for Analysis of Multi-Layered Hypersonic Thermal Protection Systems With Unknown Thermal Contact Resistances

Abstract

This work presents a systematic application of the physics-infused reduced-order modeling (PIROM) framework to the analysis of multi-layered hypersonic thermal protection systems (TPS) with unknown thermal contact resistances (TCRs).

1 Introduction

Hypersonic thermal protection systems (TPS) play a critical role in safeguarding hypersonic vehicles as they are exposed to extreme aerodynamic heating during flight. Therefore, accurate modeling and prediction of TPS behavior become essential for designing effective thermal protection strategies. Nonetheless, the highly non-linear fluid-thermal-structural interactions, temperature-dependent material property variations, and uncertainties in thermal contact resistances (TCR) at the interfaces of the TPS layers, pose significant challenges for traditional modeling approaches such as finite-element methods (FEM), as well as reduced-order (ROM) and surrogate (SM) modeling techniques.

Particularly, a critical artifact in transient reduced-order modeling of hypersonic TPS is the presence of TCRs. Due to manufacturing limitations, when two solids are brought together to form an interface, the true solid-to-solid contact area between them is generally a small fraction of the apparent area over which they meet Minges1970. The net effect of imperfect contact is the formation of a temperature discontinuity at the interface that is in general challenging to predict, and that significantly affects the heat transfer across the interface. The TCRs at the interfaces can vary by orders of magnitude depending on the surface roughness, contact pressure, and temperature [Minges1970,Deng2026]. This variability in TCRs can lead to significant uncertainties in transient temperature predictions, and thus compromise the reliability of the ROMs, for design, analysis, and optimization of hypersonic TPS [Fang2023]. Thus, there is a critical need for efficient, accurate, and generalizable modeling approaches that can effectively infer TCRs while still providing accurate predictions of TPS behavior under varying conditions such as boundary conditions and material properties.

Attempts to address the challenges of modeling hypersonic TPS with unknown TCRs have been made using various approaches, ranging from traditional physics-based modeling to data-driven techniques [X]. For instance, in Ref. [Zhang2024] a finite-element thermo-mechanical model of a three-dimensional reusable heat-pipe cooled thermal protection system is developed to perform Levenberg-Marquardt gradient-based inference of internal TCRs using boundary temperature measurements. In Ref. [X],

. . . . Other X approaches such as those in Refs.[X] . . .

While these methods offer a high-fidelity representation of the underlying physics, highly refined FEMs result in large-scale systems that are computationally expensive to solve, and thus are not suitable for design, analysis, and optimization of hypersonic TPS.

Other data-driven approaches such as those in Refs.[X] ..., leverage Bayesian Networks (BNs) to learn mappings from boundary conditions and thermocop

machine learning techniques to learn the underlying dynamics of the TPS system directly from data, while inferring the TCRs. For instance, in Refs. [X]...

While these methods offer computational efficiency, they often require large amounts of data for training, and may not generalize well to unseen conditions, such as varying boundary conditions and material properties. Moreover, they may not provide insights into the underlying physics of the system, which can be critical for design and optimization.

The objectives of this work as summarized as follows:

1. Demonstrate the systematic application of PIROM to hypersonic TPS with unknown contact resistances, emphasizing on the simultaneous inference of contact conductances and hidden dynamics under varying boundary conditions and material properties.

2. Benchmark the performance of PIROM against state-of-the-art machine learning, SciML, and traditional physics-based modeling approaches (e.g., FEM) for both inferring contact conductances and transient TPS temperature dynamics.

3. Assess the performance of the PIROM in an experimental setting,...

2 Modeling of Hypersonic Protection Systems

The following discussion introduces the governing equations in the heat conduction problem for hypersonic TPS with unknown contact resistances. Moreover, three different physics-based approaches of decreasing fidelity are presented for solving the governing equations: (i) a fine three-dimensional FEM termed the *full-order model* (FOM), (ii) a one-dimensional FEM with linear elements termed the *one-dimensional FEM* (1D-FEM), and (iii) a lumped-mass representation of the TPS termed the *lumped capacitance model* (LCM). The governing equations and their three solutions strategies are presented next.

3 Governing Equations

Consider the generic domain $\Omega \subset \mathbb{R}^d$ with $d = 1, 2, 3$ as show in Fig. X. A heat flux q_b (i.e., Neumann boundary condition) is prescribed on boundary Γ_q, while a temperature T_b (i.e., Dirichlet boundary condition) is prescribed on boundary Γ_T, and $\partial\Omega = \Gamma_q \cup \Gamma_T$. The transient heat conduction dynamics is governed by the following energy equation

4 Full-Order Model

5 One-Dimensional Model

This approach follows the classical FEM approach with a linear basis function to capture the spatially varying temperature distributions inside each component of the TPS. As with the FOM, the TCR introduces additional considerations in the formulation of the 1D-FEM.

6 Lumped Capacitance Model

The lumped capacitance model (LCM) is derived from an energy conservation standpoint.

Consider the lumped-mass average temperature state,

$$\mathbf{u} = [u_1, u_2, u_3, u_4]^T \in \mathbb{R}^N \tag{1}$$

and temperature-dependent lumped-capacitance matrix,

$$A(\mathbf{u}) = \begin{bmatrix} a_1(u_1) & 0 & 0 & 0 \\ 0 & a_2(u_2) & 0 & 0 \\ 0 & 0 & a_3(u_3) & 0 \\ 0 & 0 & 0 & a_4(u_4) \end{bmatrix} \tag{2}$$

Let the external Neumann forcing be applied to the first lumped mass,

$$\mathbf{f}(t) = \begin{bmatrix} f(t) \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{3}$$

For interfaces $i = 1, 2, 3$, define the known *bulk resistances*

$$S_i(\mathbf{u}) = R_i^-(\mathbf{u}) + R_i^+(\mathbf{u}) \tag{4}$$

and the unknown contact resistance,

$$R_{xi} > 0, \qquad g_{xi} := \frac{1}{R_{xi}}. \tag{5}$$

The classical lumped-capacitance model with effective series resistances is,

$$A(\mathbf{u})\,\dot{\mathbf{u}} = B(\mathbf{u})\,\mathbf{u} + \mathbf{f}(t) \tag{6}$$

where,

$$B(\mathbf{u}) = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u}) + R_{xi}} \mathbf{K}_i \tag{7}$$

with standard nearest-neighbor diffusion stencils,

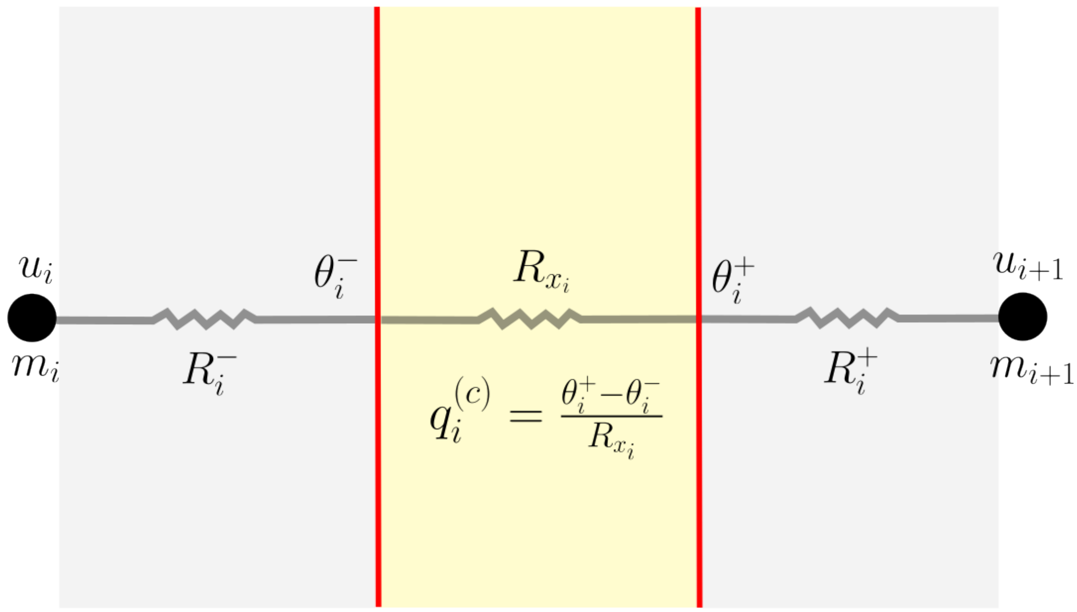
$$\mathbf{K}_1 = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \mathbf{K}_3 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \tag{8}$$

The effective heat flux through interface i is therefore,

$$q_i = \frac{u_{i+1} - u_i}{S_i(\mathbf{u}) + R_{xi}} \tag{9}$$

where the temperature-drop convention is always taken as,

$$\Delta u_i = u_{i+1} - u_i \tag{10}$$



7 Total drop = bulk drop + contact drop

To separate bulk and contact contributions, introduce interface temperatures

$$\theta_i^+, \qquad \theta_i^-, \qquad i = 1, 2, 3, \tag{11}$$

where θ_i^- and θ_i^+ are the temperatures immediately to the left and right of contact i, respectively. Define the contact temperature jumps as,

$$\Delta\theta_i := \theta_i^+ - \theta_i^-, \qquad i = 1, 2, 3 \tag{12}$$

The total lumped-to-lumped temperature drop across interface i,

$$u_{i+1} - u_i \tag{13}$$

This total lumped-to-lumped temperature drop is the sum of the bulk drop across the interface and the contact jump across the interface,

$$\underbrace{u_{i+1} - u_i}_{\text{total drop}} = \underbrace{\left(\theta_i^- - u_i\right) + \left(u_{i+1} - \theta_i^+\right)}_{\text{bulk drop}} + \underbrace{\left(\theta_i^+ - \theta_i^-\right)}_{\text{contact drop}}. \tag{14}$$

Hence,

$$\underbrace{\left(\theta_i^- - u_i\right) + \left(u_{i+1} - \theta_i^+\right)}_{\text{bulk drop}} = \underbrace{\left(u_{i+1} - u_i\right)}_{\text{total drop}} - \underbrace{\Delta\theta_i}_{\text{contact drop}} \tag{15}$$

$$\text{bulk drop across interface } i = (u_{i+1} - u_i) - \Delta\theta_i. \tag{16}$$

Thus the lifted state is

$$x = \begin{bmatrix} u \\ \Delta\boldsymbol{\theta} \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ \Delta\theta_1 \\ \Delta\theta_2 \\ \Delta\theta_3 \end{bmatrix} \in \mathbb{R}^7, \qquad \Delta\boldsymbol{\theta} = \begin{bmatrix} \Delta\theta_1 \\ \Delta\theta_2 \\ \Delta\theta_3 \end{bmatrix}.$$

8 Heat-flux Continuity and DAE Formulation

Using the adopted convention, define the bulk and contact heat fluxes as

$$q_i^{(b)} = \frac{(u_{i+1} - u_i) - \Delta\theta_i}{S_i(u)}, \qquad q_i^{(c)} = \frac{\Delta\theta_i}{R_{xi}} = g_{xi}\Delta\theta_i. \tag{17}$$

Heat-flux continuity across each interface requires

$$q_i^{(b)} = q_i^{(c)}, \qquad i = 1, 2, 3, \tag{18}$$

which yields the algebraic constraints

$$\frac{(u_{i+1} - u_i) - \Delta\theta_i}{S_i(u)} - g_{xi}\Delta\theta_i = 0, \qquad i = 1, 2, 3. \tag{19}$$

Lumped-temperature dynamics

The four lumped energy balances yield,

$$a_1(u_1)\dot{u}_1 = f(t) - q_{12}^{(b)} \tag{20a}$$
$$a_2(u_2)\dot{u}_2 = q_{12}^{(b)} - q_{23}^{(b)} \tag{20b}$$
$$a_3(u_3)\dot{u}_3 = q_{23}^{(b)} - q_{34}^{(b)} \tag{20c}$$
$$a_4(u_4)\dot{u}_4 = q_{34}^{(b)} \tag{20d}$$

Substituting using the bulk heat fluxes yields,

$$a_1(u_1)\dot{u}_1 = f(t) - \frac{(u_2 - u_1) - \Delta\theta_1}{S_1(u)} \tag{21a}$$
$$a_2(u_2)\dot{u}_2 = \frac{(u_2 - u_1) - \Delta\theta_1}{S_1(u)} - \frac{(u_3 - u_2) - \Delta\theta_2}{S_2(u)} \tag{21b}$$
$$a_3(u_3)\dot{u}_3 = \frac{(u_3 - u_2) - \Delta\theta_2}{S_2(u)} - \frac{(u_4 - u_3) - \Delta\theta_3}{S_3(u)} \tag{21c}$$
$$a_4(u_4)\dot{u}_4 = \frac{(u_4 - u_3) - \Delta\theta_3}{S_3(u)} \tag{21d}$$

Matrix form of DAE

Define the difference matrix,

$$\mathbf{D} = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}, \ \mathbf{D}u = \begin{bmatrix} u_2 - u_1 \\ u_3 - u_2 \\ u_4 - u_3 \end{bmatrix} \tag{22}$$

Define the bulk temperature stencils,

$$\mathbf{K}_1 = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \ \mathbf{K}_2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \ \mathbf{K}_3 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \tag{23}$$

and the bulk jump-coupling stencils,

$$\mathbf{L}_1 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \ \mathbf{L}_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \ \mathbf{L}_3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{bmatrix} \tag{24}$$

Then,

$$A(u)\dot{u} = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \Big(\mathbf{K}_i u + \mathbf{L}_i \Delta\boldsymbol{\theta} \Big) + \mathbf{f}(t) \tag{25}$$

and the algebraic constraints for heat flux continuity are written as,

$$\mathrm{diag}\left(\frac{1}{S_1(u)}, \frac{1}{S_2(u)}, \frac{1}{S_3(u)} \right) \mathbf{D}u - \left[\mathrm{diag}\left(\frac{1}{S_1(u)}, \frac{1}{S_2(u)}, \frac{1}{S_3(u)} \right) + G_x \right] \Delta\boldsymbol{\theta} = 0 \tag{26}$$

where

$$G_x = \begin{bmatrix} g_{x1} & 0 & 0 \\ 0 & g_{x2} & 0 \\ 0 & 0 & g_{x3} \end{bmatrix} \tag{27}$$

Equation (26), along with the lumped-temperature dynamics, defines a DAE system for the lifted state $x = [u, \Delta\boldsymbol{\theta}]$.

9 DAE index reduction

The DAE is index-reduced by differentiating the heat-flux continuity constraints with respect to time, which, for each interface i, yields,

$$\frac{d}{dt} \left[\frac{(u_{i+1} - u_i) - \Delta\theta_i}{S_i(u)} - g_{xi}\Delta\theta_i \right] = 0 \tag{28}$$

The contact conductances g_{xi} are assumed constant, therefore,

$$\big(1 + g_{xi} S_i(u)\big)\Delta\dot{\theta}_i = \big(\dot{u}_{i+1} - \dot{u}_i\big) - \frac{(u_{i+1} - u_i) - \Delta\theta_i}{S_i(u)} \dot{S}_i(u) \tag{29}$$

Using the original algebraic constraint

$$\frac{(u_{i+1} - u_i) - \Delta\theta_i}{S_i(u)} = g_{xi}\Delta\theta_i, \tag{30}$$

the index-reduced dynamics become,

$$\boxed{\Delta\dot{\theta}_i = \frac{\dot{u}_{i+1} - \dot{u}_i - g_{xi}\Delta\theta_i\,\dot{S}_i(u)}{1 + g_{xi}S_i(u)}, \qquad i = 1, 2, 3} \tag{31}$$

Define

$$M(u; R_x) = \begin{bmatrix} 1 + g_{x1}S_1(u) & 0 & 0 \\ 0 & 1 + g_{x2}S_2(u) & 0 \\ 0 & 0 & 1 + g_{x3}S_3(u) \end{bmatrix} \tag{32}$$

and,

$$\dot{S}(u) = \begin{bmatrix} \dot{S}_1(u) \\ \dot{S}_2(u) \\ \dot{S}_3(u) \end{bmatrix}, \qquad \Delta\boldsymbol{\theta} \odot \dot{S}(u) = \begin{bmatrix} \Delta\theta_1\dot{S}_1(u) \\ \Delta\theta_2\dot{S}_2(u) \\ \Delta\theta_3\dot{S}_3(u) \end{bmatrix} \tag{33}$$

Then, the index-reduced dynamics for the contact jumps can be written in matrix form as,

$$M(u; R_x)\Delta\dot{\theta} = D\,\dot{u} - G_x\big(\Delta\boldsymbol{\theta} \odot \dot{S}(u)\big) \tag{34}$$

Thus, the index-reduced lifted system is given by the coupled ODEs,

$$A(u)\dot{u} = \sum_{i=1}^{N-1} \frac{1}{S_i(u)}\Big(\mathbf{K}_i u + \mathbf{L}_i\Delta\boldsymbol{\theta}\Big) + \mathbf{f}(t) \tag{35a}$$

$$M(u; R_x)\Delta\dot{\theta} = \mathbf{D}\dot{u} - G_x\big(\Delta\boldsymbol{\theta} \odot \dot{S}(u)\big) \tag{35b}$$

10 Physics-Infused Reduced-Order Modeling

The formulation of PIROM for TPS starts by mathematically connecting the FOM, i.e., the DG-FEM, to the RPM, i.e., the LCM. This results presented in this section leverage the previous work in Ref. X, where a coarse-graining procedure is developed to systematically derive the LCM from the DG-FEM. Such coarse-graining procedure pinpoints the missing dynamics in the LCM when compared to DG-FEM, enabling the systematic design of a memory closure to account for residual dynamics associated with discarded spatial resolution.

11 Bulk-only Mori–Zwanzig Memory Closure

The purpose of the closure is to account for the unresolved dynamics in the LCM, while leaving the contact-jump dynamics unchanged. Accordingly, the

Accordingly, write the Projected Generalized Langevin Equation (PGLE) for the lumped-temperature dynamics,

$$A(\mathbf{u})\dot{\mathbf{u}} = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u})}\left(\mathbf{K}_i u + \mathbf{L}_i\Delta\boldsymbol{\theta}\right) + \mathbf{f}(t) + \int_{t_0}^{t} \boldsymbol{\kappa}(t, s, u, \mathbf{f}; \Theta)ds \tag{36}$$

Now, ensure the designed memory modifies only the bulk conductances, i.e.,

$$\boldsymbol{\kappa} \in \operatorname{span}\left\{\mathbf{K}_1 u(t), \mathbf{K}_2 u(t), \mathbf{K}_3 u(t), \mathbf{L}_1 \Delta\boldsymbol{\theta}_1(t), \mathbf{L}_2 \Delta\boldsymbol{\theta}_2(t), \mathbf{L}_3 \Delta\boldsymbol{\theta}_3(t)\right\} \tag{37}$$

which leads to,

$$\boldsymbol{\kappa}(t, s, u, \mathbf{f}; \Theta) \approx \sum_{j=1}^{m} \kappa_j(t, s, u, \mathbf{f}; \Theta) \boldsymbol{\mathcal{P}}_j \tag{38}$$

where,

$$\boldsymbol{\mathcal{P}}_j = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \mathbf{p}_{ij} \tag{39}$$

Now, define the memory variables as,

$$\beta_j(t) = \int_{t_0}^{t} \kappa_j(t, s, u, \mathbf{f}; \Theta) ds \tag{40}$$

then,

$$\int_{t_0}^{t} \boldsymbol{\kappa}(t, s, u, \mathbf{f}; \Theta) ds = \sum_{j=1}^{m} \beta_j(t) \boldsymbol{\mathcal{P}}_j \tag{41}$$

It follows that,

$$\sum_{j=1}^{m} \beta_j(t) \boldsymbol{\mathcal{P}}_j = \sum_{j=1}^{m} \beta_j(t) \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \mathbf{p}_{ij} = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \left(\sum_{j=1}^{m} \mathbf{p}_{ij} \beta_j(t) \right) = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \mathbf{P}_i \boldsymbol{\beta} \tag{42}$$

Then, the memory correction to the LCM dynamics leads to the following modified lumped-temperature dynamics,

$$A(\mathbf{u})\dot{\mathbf{u}} = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u})} \left(\mathbf{K}_i u + \mathbf{L}_i \Delta\boldsymbol{\theta} + \mathbf{P}_i \boldsymbol{\beta} \right) + \mathbf{f}(t) \tag{43}$$

Now, let,

$$\kappa_i\left(t, s, \mathbf{u}, \mathbf{f}; \Theta\right) = e^{-\lambda_j(t-s)} \left(q_j^\top \phi(\mathbf{u}(s)) + r_j^\top \psi(\mathbf{f}(s)) \right) \tag{44}$$

such that the memory variables are given by,

$$\boldsymbol{\beta}(t) = \int_{t_0}^{t} e^{-\Lambda(t-s)} \left(Q^\top \phi(\mathbf{u}(s)) + R^\top \psi(\mathbf{f}(s)) \right) ds \tag{45}$$

The hidden dynamics are now given by,

$$\dot{\boldsymbol{\beta}} = -\Lambda \boldsymbol{\beta} + Q^\top \phi(\mathbf{u}) + R^\top \psi(\mathbf{f}) \tag{46}$$

and in state-space form, the hidden dynamics are,

$$\dot{\boldsymbol{\beta}} = -\Lambda \boldsymbol{\beta} + Q^\top \phi(\mathbf{u}) + R^\top \psi(\mathbf{f}) \tag{47}$$

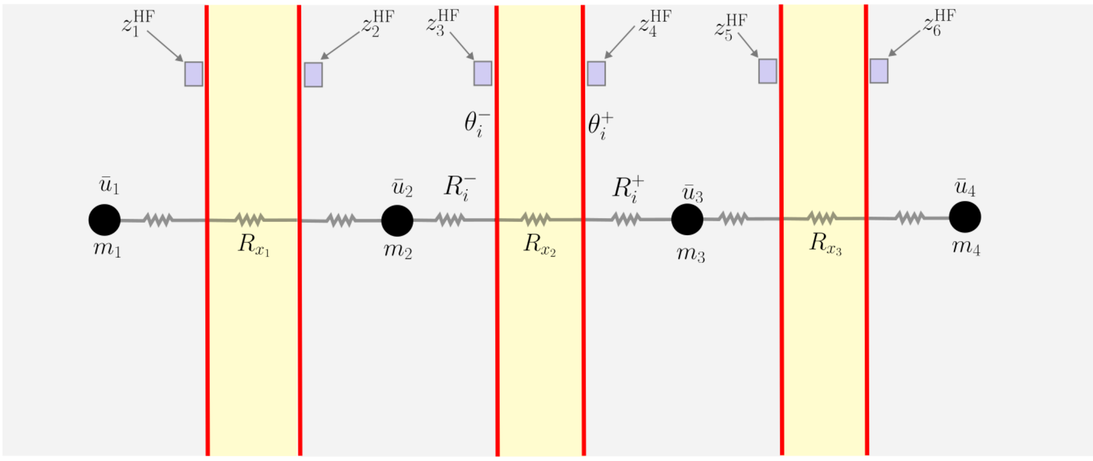
The full lifted and closed system is given by,

$$A(\mathbf{u})\dot{\mathbf{u}} = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u})} \left(\mathbf{K}_i \mathbf{u} + \mathbf{L}_i \Delta\boldsymbol{\theta} + \mathbf{P}_i \boldsymbol{\beta} \right) + \mathbf{f}(t) \tag{48a}$$

$$M(\mathbf{u}; R_x)\dot{\Delta\boldsymbol{\theta}} = \mathbf{D}\dot{\mathbf{u}} - G_x\big(\Delta\boldsymbol{\theta} \odot \dot{S}(\mathbf{u})\big) \tag{48b}$$

$$\dot{\boldsymbol{\beta}} = -\Lambda \boldsymbol{\beta} + Q^\top \phi(\mathbf{u}) + R^\top \psi(\mathbf{f}) \tag{48c}$$

The main conclusion of this work are as follows,



$$q_i^{(c)} = \frac{\theta_i^+ - \theta_i^-}{R_{x_i}}$$

1. The PIROM framework successfully learns a ROM for the prediction of the temperature dynamics of the TPS, while simultaneously inferring the unknown contact conductances with errors of less than X% on noisy data with SNR of up to Y dB.

2. The PIROM outperforms the state-of-the-art data-driven, NN-based SciML, and traditional FEM modeling approaches for inferring contact conductances, as well as for predicting the temperature dynamics for OOD configurations.

3. The PIROM is successfully deployed in an experimental setting, where it is able to infer the contact conductances and predict the temperature dynamics with errors of less than X% on noisy data with SNR of up to Y dB.